

Monte Carlo Counterparty Credit Risk Engine

Executive summary: what the risk is, how we measure it, what we found

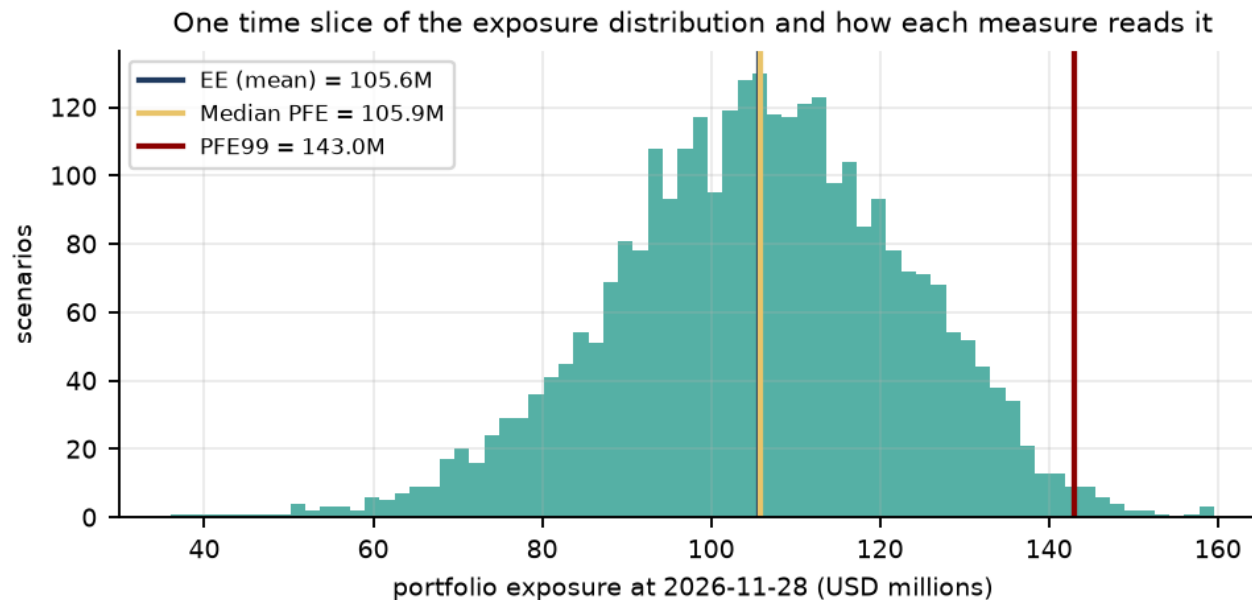
ESF derivatives book | 16 trades | 3 counterparties | PFE at the 99th percentile | full detail in [Capitolis_CCR_Complete_Report.pdf](#)

The answer in one page

\$108M	\$122M	\$175M	99%
exposure if everyone defaulted today	peak expected exposure (EE)	peak PFE99 (MPE), 2026-10-09	of today's exposure is CPTY_C

- Uncollateralized book: our credit risk is short-dated. Most exposure disappears by December 2026 as trades mature.
- Concentrated: one \$500M bond forward (BF_0003) with CPTY_C is almost all of it. A limit or margin on that trade moves the answer more than any model choice.
- The three measures together matter: at peak, median PFE \$121.3M, EE \$121.9M, PFE99 \$174.7M.
- If any trade is actually margined, exposure falls sharply (collateral slide). Confirming CSA terms is the most valuable open question.

What we measure



Exposure = $\max(\text{net value}, 0)$: only gains against us are at risk. EE is the average; the median PFE is the typical case; PFE99 is the level exceeded in only 1 of 100 scenarios; MPE is the peak PFE over time.

The book: 16 trades, short-dated

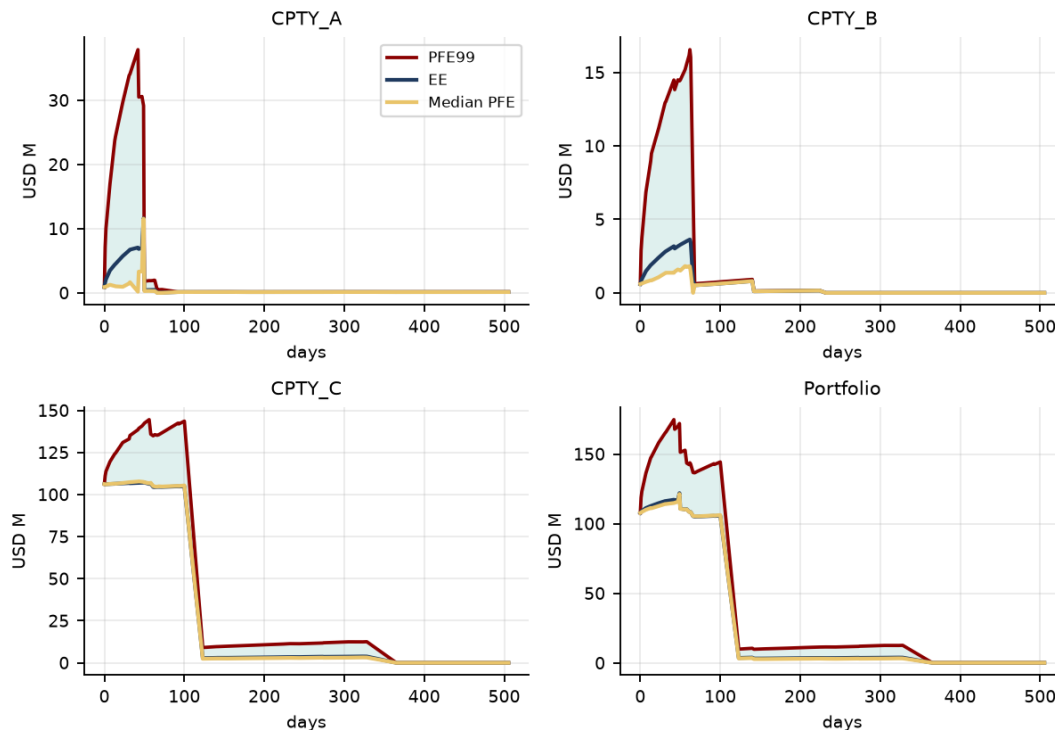


8 equity total return swaps (incl. JPY compo), 4 bond forwards, 4 bond TRS across three counterparties. All priced with the supplied, independently reviewed pricer library.

How the engine works

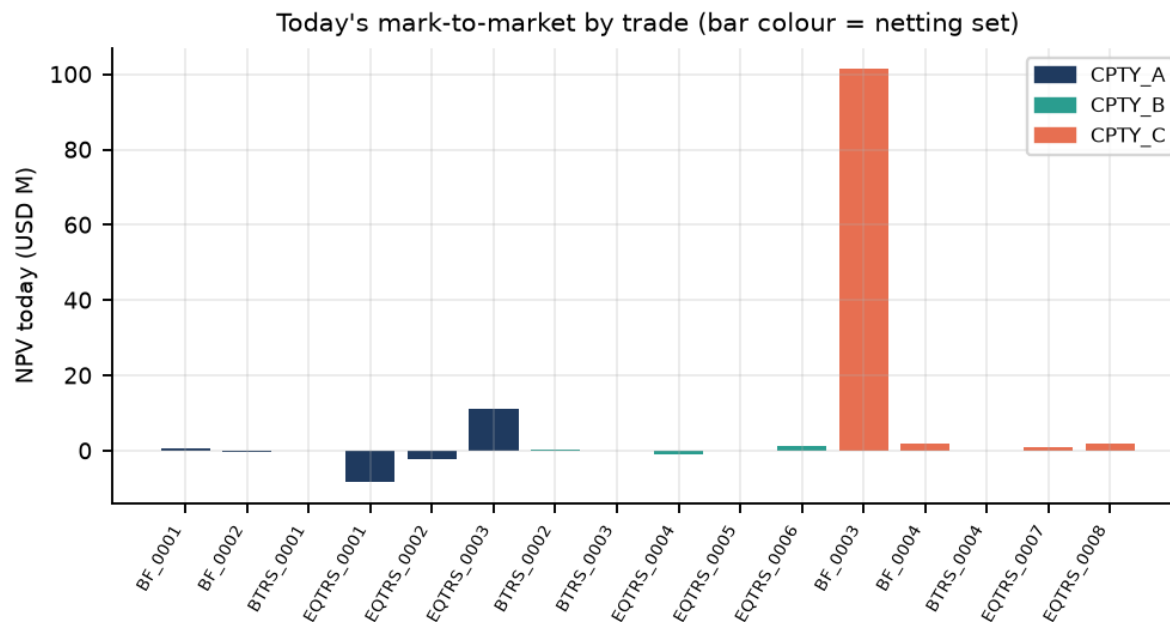
Step	What we do	Choice
1 Calibrate	SOFR curve from CME futures; realised vols; 39x39 correlation; mean reversion from swaptions	Real market data throughout
2 Simulate	USD rate: Hull-White one-factor. Equities and USDJPY: correlated GBM driven by the rate	Exact fit to today's curve; negative-rate capable
3 Sample	Latin-Hypercube scenarios (5,000 recommended for reporting; 3,000 used here); Cholesky for correlation (PCA factor model optional)	Lowest measured error of 5 methods
4 Dates	Market pillar dates (O/N..10Y) plus every trade's own reset and maturity date	Industry convention, exact cash-flow dates
5 Reprice	Every trade re-priced in every scenario and date with the standard pricers	No new pricing logic
6 Aggregate	Net by counterparty, $\max(.,0)$, then EE, median PFE, PFE99, MPE	Uncollateralized by default

Exposure through time



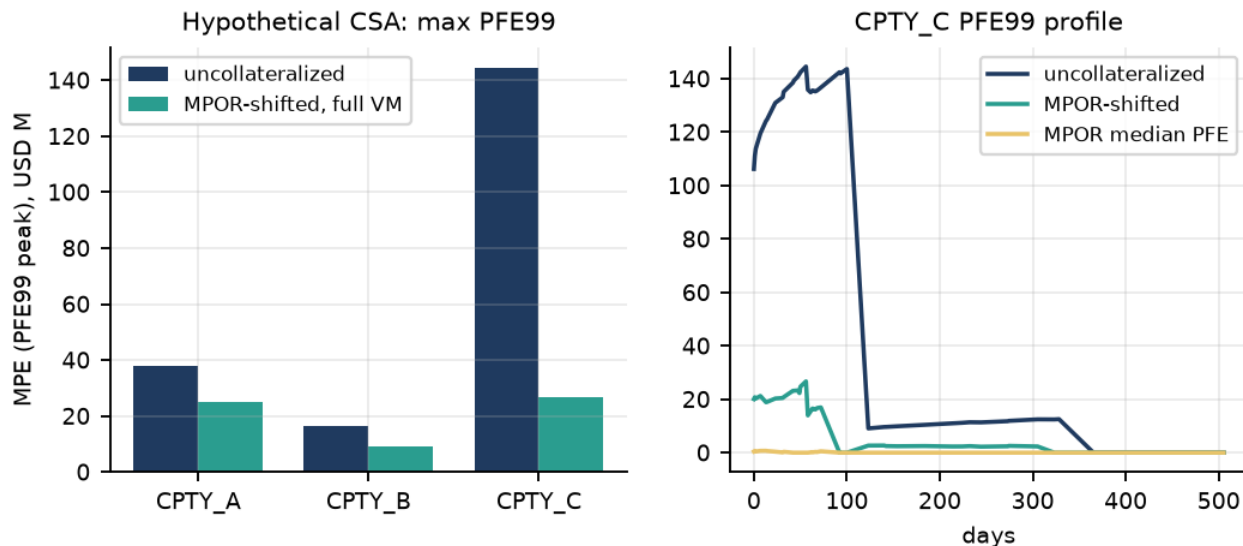
Exposure peaks within about 2 months and collapses as trades mature; a small tail runs to January 2028.

Where the risk sits



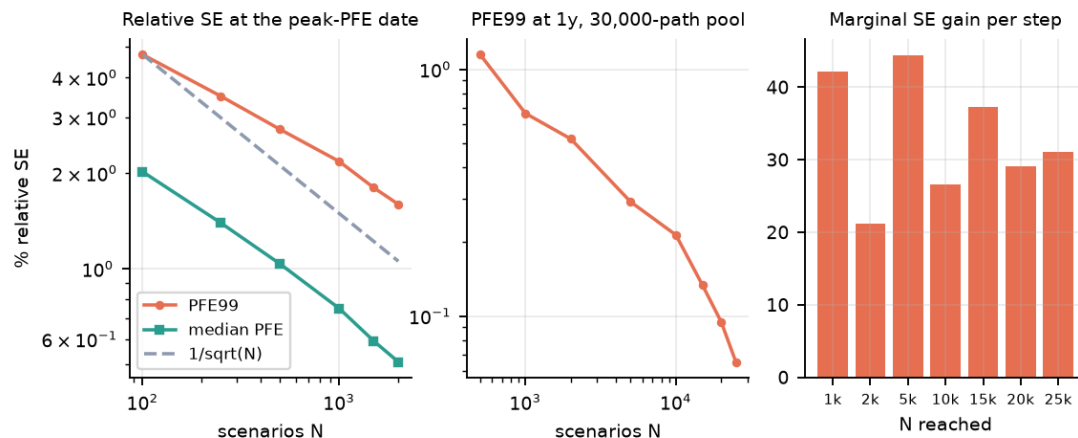
CPTY_C's value is one trade (BF_0003, +\$101M). CPTY_B nets negative so has zero exposure today. This is concentration risk, not diversified counterparty risk.

What if margin were posted? (hypothetical)



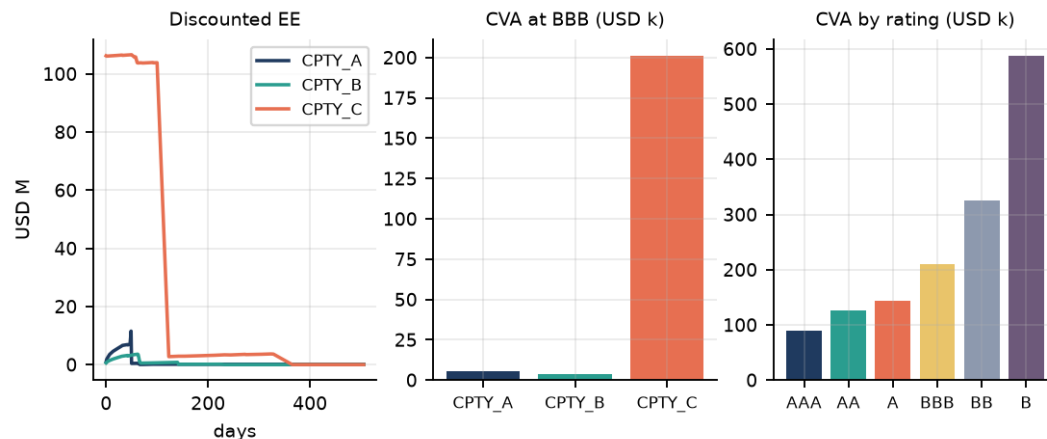
Full variation margin with a 10-day margin period of risk: peak PFE99 falls CPTY_A 34%, CPTY_B 45%, CPTY_C 82%. No CSA data exists for the real book, so this illustrates a built capability.

How precise, and how do we know it is right



- Decision: Latin Hypercube sampling: 51x lower error than pseudo-random on the controlled test, and better on both PFE99 (1.15x) and median PFE (1.57x) on the real engine.
- Decision: 5,000 paths for standard PFE99 reporting (error 1.0% at the worst-case date, 0.29% at 1 year); 1,000 for iteration; 10,000 for limit sign-off. This deck uses 3,000.
- Checks passed: t=0 self-consistency 0.0000%, martingale test (max 0.73 bp), parametric VaR ratio 1.08, stress-test directions, 76 automated tests.
- Model risk (mean reversion, volatility proxy) outweighs sampling noise beyond about 10,000 paths.

Credit valuation adjustment and Basel SA-CVA capital



- CVA (unilateral, uncollateralized, 60% LGD, BBB bond-spread proxy): \$210,394 in total, 96% from CPTY_C; it ranges from \$126k (AA) to \$325k (BB) so the rating assumption matters.
- SA-CVA capital \$1.93M (RWA \$24.1M), 90% of it counterparty credit spread risk; interest rate, FX and equity classes are small. Sensitivities by common-random-number bumps, Basel MAR50 parameters.
- Assumed: counterparty ratings, bond-index spread proxy, no wrong-way risk, no collateral. Needs supervisory approval for regulatory use.

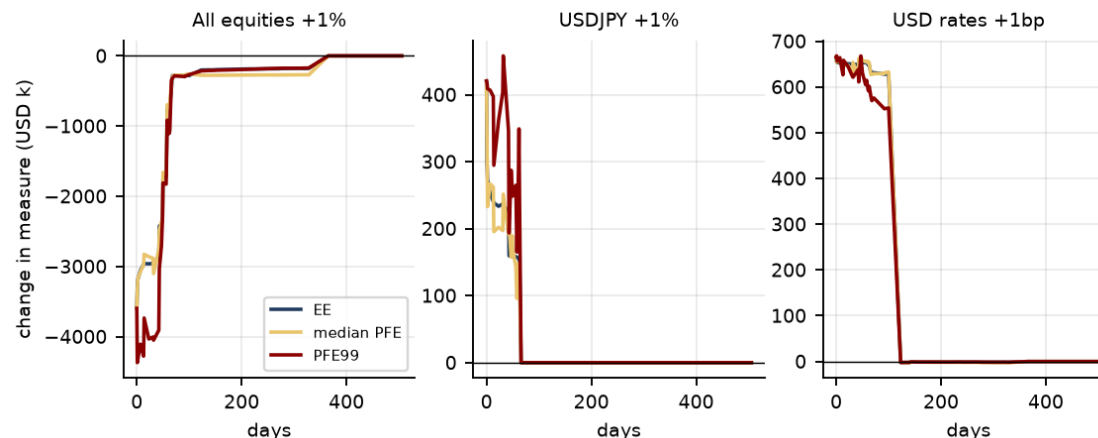
Data sources

Input	Source	Used for
Trades, pricing library	Capitolis (supplied)	16 trades; all valuation
USD SOFR futures	CME via Databento	USD curve (our bootstrap)
Equities, USDJPY, sectors	yfinance	Spots, vols, correlation, SA-CVA buckets
SOFR history; credit spreads by rating	FRED (SOFR, ICE BofA OAS)	Rate vol; CVA credit proxy
JPY OIS history (35 tenors, 2011-2026)	Bloomberg file from the project team (data/raw/sources)	JPY curve, vol, differential, mean-reversion test
TONA; JGB yields	Bank of Japan API; Japan MoF	JPY rate correlation; cross-checks
Swaption cubes, FX forwards	Bloomberg snapshot (licensed)	USD mean reversion, FX forwards
SA-CVA rules	BIS Basel MAR50 (2020)	Risk weights, correlations, aggregation

Methodology at a glance

Component	Method	Key choice
USD rates	Hull-White one-factor	$a = 0.0167$ (swaptions), $\sigma = 0.63\%$ (SOFR)
Equities, FX	Correlated GBM	3y realised vols, static 39x39 correlation, Cholesky
JPY	Second Hull-White factor, negative-rate capable	Curve and vol from the daily JPY OIS history; mean reversion at lower bound 0.001
Sampling	Latin Hypercube	5,000 paths reporting; 3,000 here
Dates	Market pillars + trade events	42 nodes
Exposure	Net, $\max(V, 0)$; EE, median PFE, PFE99, MPE	Uncollateralized; MPOR 10 business days as hypothetical
CVA	Regulatory CVA (MAR50.32)	BBB bond-spread proxy, LGD 60%, unilateral
SA-CVA	Bump sensitivities, MAR50 aggregation	Common random numbers, 21 runs, $m_CVA = 1$

Greeks: book and exposure sensitivities



- Complete set, by netting set: t=0 delta, gamma, DV01 (parallel and 8 buckets), FX; and the sensitivities of EE, median PFE and PFE99 to all 37 equities, USDJPY, USD rates and volatilities.
- At peak PFE99 (2026-10-09): +1% on all equities moves PFE99 by \$-3,908k and EE by \$-2,869k (pay-equity swaps: falling shares raise our claim); +1bp on USD rates moves PFE99 by +\$638k.
- Method: bump-and-reprice with common random numbers; equity and FX bumps reuse the simulated paths (exact GBM rescaling, only affected trades repriced), so 82 bumps cost a fraction of re-simulation.
- Validated against analytic t=0 deltas, bump-size stability, additivity of single-name deltas, and common random numbers versus independent draws.

Assumptions, limits, next steps

- Volatility is a 3-year realised proxy (no single-name options data); one static correlation matrix; GBM understates fat tails.
- JPY: a real negative-rate-capable Hull-White factor is built (curve and volatility from the daily JPY OIS history; USD-JPY rate correlation calibrated near zero) but does not yet drive JPY equity drift; mean reversion is at its lower bound because JPY vol rises with tenor in three independent datasets.
- No counterparty default probability or wrong-way risk: this is exposure, not expected loss.
- Ask: are any trades margined, and under what CSA terms?
- Next: wire JPY factor into the drift, two-factor rate model, model-risk study on mean reversion, vols and correlation.